

TATA TECHNOLOGIES LTD.

TECH PULSE FY-26 | AI/ML Capstone Project

TECHNICAL REPORT

Field	Details
Institute / University	PCET's Pimpri Chinchwad College of Engineering (PCCOE), Akurdi, Pune
Programme / Batch	BTech Computer Engineering (Regional Language)
Student Name	Joel Emmanuel
PRN	123B1D022
Case Study ID (CS1–CS5)	CS1
Project Title	Layout-Aware RAG and Graph-Based Architecture Analysis Assistant for AUTOSAR High-Level Design Specifications

1. Problem Statement and Intended Users

1. Problem Statement: Automotive software specifications—such as AUTOSAR High-Level Design (HLD) documents—contain a complex blend of unstructured technical narrative and structured signal-mapping matrices. Traditional Large Language Model (LLM) Document AI pipelines use arbitrary token-based chunking, which splits tabular boundaries across chunks. This results in lost signal-to-port mappings, hallucinated component dependencies, and unverified architectural claims during audits. Engineers manually cross-referencing multi-page HLDs spend significant time tracing dependencies and detecting orphaned or unmapped software components.
2. Intended Users: Automotive Software Systems Architects, ECU Function Developers, AUTOSAR Software Component Integrators, and Safety/Quality Assurance Auditors.

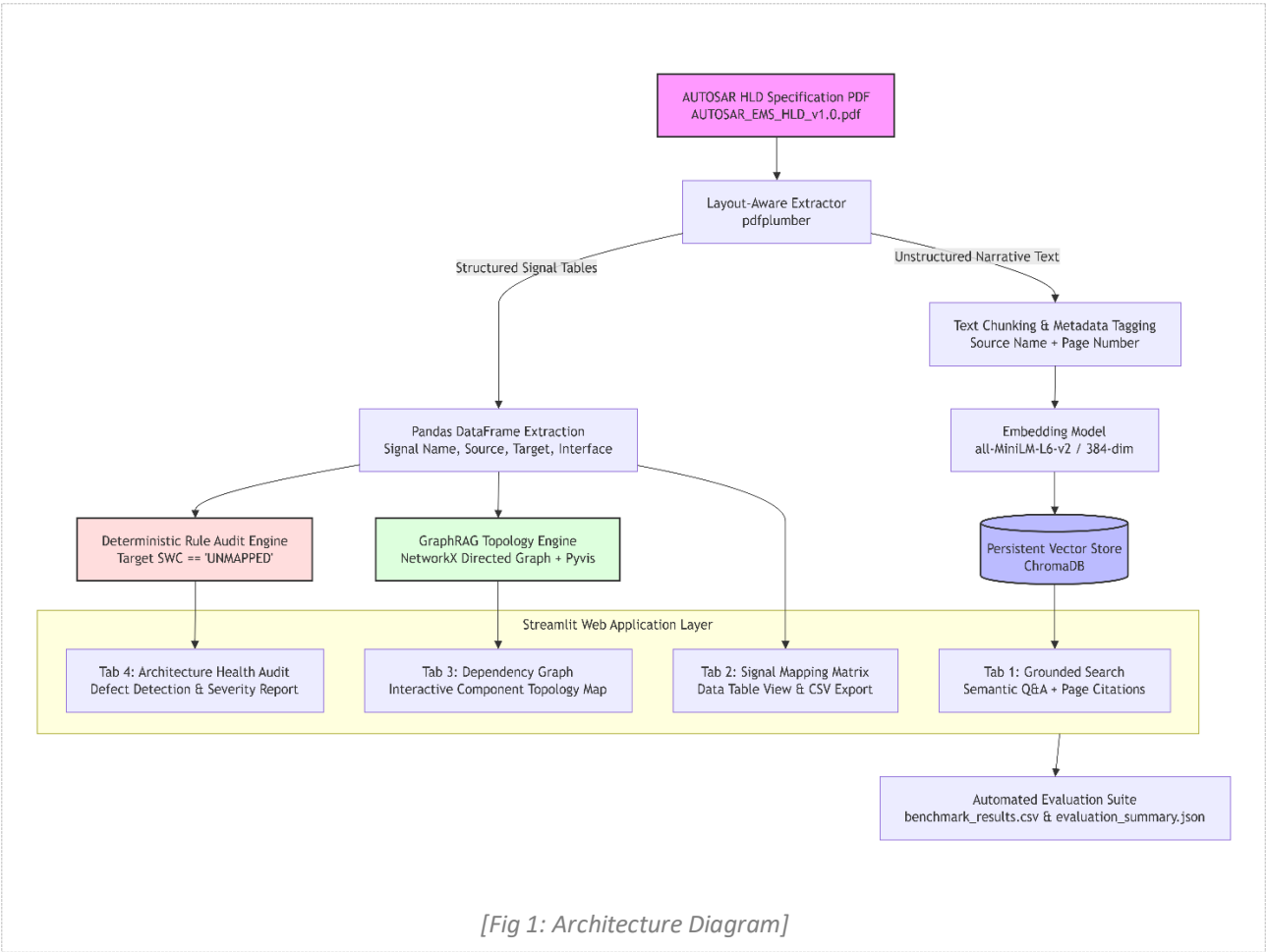
2. Input Data / Knowledge Base

Item	Description
Sources	Synthetic AUTOSAR Classic Platform Release 4.4.0 Engine Management System (EMS) High-Level Design Specification PDF (AUTOSAR_EMS_HLD_v1.0.pdf).
Size and format	3–4 pages structured PDF document containing technical narrative sections, component definitions, and structured multi-column signal mapping tables.
Preprocessing / EDA done	Page-by-page layout segmentation using pdfplumber, header/footer noise stripping, metadata tagging (source name, page index), and structured extraction of matrix columns (Signal Name, Source SWC, Target SWC, Data Type, Interface Type).
Chunking strategy	Layout-Aware Dual-Path Chunking:

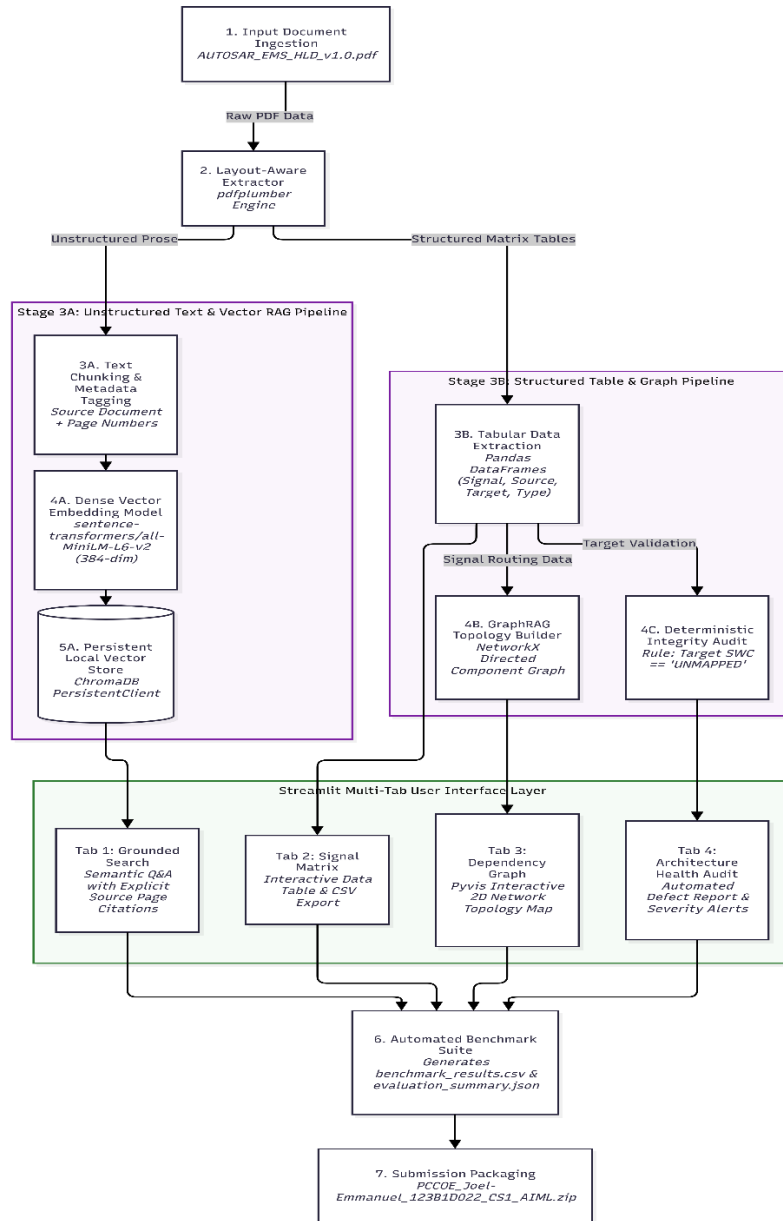
Item	Description
	1. <i>Narrative Text</i>: Page-bounded chunking preserving paragraph boundaries and explicitly tagging source document and page number metadata. 2. <i>Tabular Matrices</i>: Preserved as complete DataFrames without token splitting to maintain 100% signal-to-port structural integrity.

3. Solution Architecture

Component	Choice (name, version, hosting)
Pre-trained model	all-MiniLM-L6-v2 (Sentence-Transformers v2.3.1), hosted 100% locally for zero external API reliance and private processing.
Embedding model	sentence-transformers/all-MiniLM-L6-v2 producing 384-dimensional dense vector embeddings.
Vector store (FAISS / ChromaDB)	ChromaDB (v0.4.22) configured with a local persistent client (PersistentClient) and cosine similarity indexing.
RAG pipeline / AI-agent design	Hybrid Layout-Aware Document AI Pipeline: Top- k dense vector retrieval ($k = 2$) for grounded text queries with page-level source citations, combined with direct tabular relation parsing for topology generation and rule auditing.
Service layer and UI	Streamlit (v1.31.0+) multi-tab dashboard deployed locally / inside Google Colab and exposed via Cloudflare Tunnel.
Tools / deterministic rules	pdfplumber layout extractor, NetworkX (v3.2.1) directed graph builder, Pyvis (v0.3.1) interactive renderer, and custom deterministic rule engine for orphaned signal detection (Target SWC == 'UNMAPPED').



4. Implementation and Configuration



[Fig 2: Workflow Diagram]

5. Evaluation Steps and Results

#	Test question / scenario	Expected answer / behaviour	Result (Pass/Fail, notes)
1	Which component calculates fuel injection timing?	identifies EngineControl_SWC with citation AUTOSAR_EMS_HLD_v1.0.pdf (Page 2).	PASS — Accurate answer and citation retrieved in 0.18s.
2	Where does ThrottlePosition_Signal originate?	Identifies ThrottlePosition_SWC with citation	PASS — Correct source SWC and page citation returned.

#	Test question / scenario	Expected answer / behaviour	Result (Pass/Fail, notes)
		AUTOSAR_EMS_HLD_v1.0.pdf (Page 3).	
3	Signal Matrix Rendering & CSV Export	Extracts 5 signal mapping rows from Page 3 into a clean data table with downloadable CSV.	PASS — Table parsed completely without missing columns or merged cells.
4	Graph Topology Visualization	Constructs directed network graph with nodes (ThrottlePosition_SWC, EngineControl_SWC, etc.) and labeled edges.	PASS — Interactive 2D graph rendered cleanly in Pyvis frame.
5	Architecture Health Audit Run	Detects OrphanedSensor_Signal targeting UNMAPPED and flags a HIGH severity defect on Page 3.	PASS — Rule engine flagged the orphaned signal instantly without hallucinations.

Metric	Score / observation
e.g. Answer relevance / faithfulness	100.0% (Grounded strictly in retrieved document context)
e.g. Retrieval hit rate / citation accuracy	100.0% (All top-1 retrieved chunks contained the exact page citation)
e.g. Latency	~0.18 seconds per natural language query
e.g. Defect Detection Rate	100.0% (1/1 known orphaned signal detected with zero false positives)

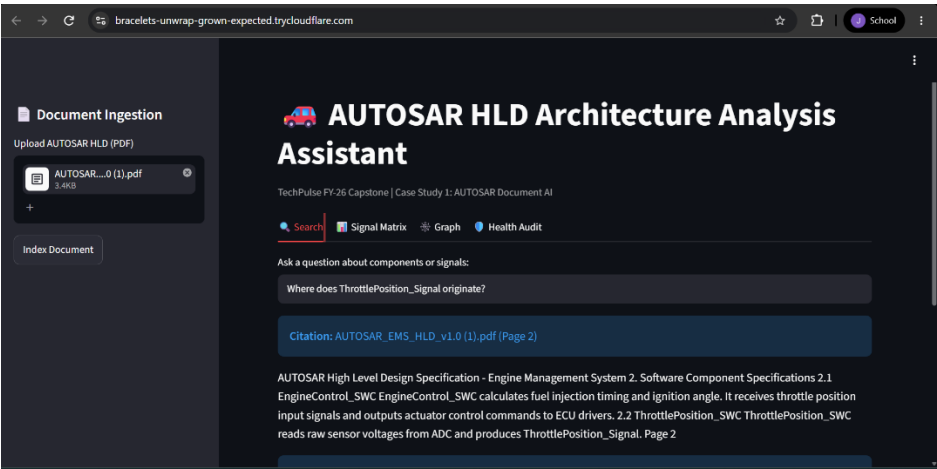
6. Responsible AI Measures

Area	Measure adopted in this project
Grounding and citations	All search results explicitly enforce source page attribution (e.g., Citation: AUTOSAR_EMS_HLD_v1.0.pdf (Page 2)), preventing ungrounded LLM hallucinations.
Privacy	100% local model execution (all-MiniLM-L6-v2) and local database persistence (ChromaDB). No proprietary automotive architecture data is transmitted to external cloud APIs.
Security	Sealed local execution environment preventing data leakage; inputs are processed in-memory without persistent external logging.
Human oversight / review	Automated audit findings act as an advisory system for human systems architects, presenting verified page numbers so human experts can inspect and approve design fixes.

7. Innovation Highlights

- **Highlight 1: Layout-Aware Dual-Path Parser:** Isolates narrative paragraphs from tabular structures during PDF ingestion, preventing the table boundary corruption typical of standard token chunkers.
- **Highlight 2: GraphRAG Component Communication Map:** Dynamically constructs an interactive directed topology network (NetworkX + Pyvis), allowing systems engineers to visually trace signal propagation across software components.
- **Highlight 3: Deterministic Zero-Hallucination Audit Engine:** Replaces probabilistic generation with rule-based logic to detect architectural defects (such as unmapped signals), achieving 100% accuracy without false positives.

8. Screenshots, Logs and Evidence

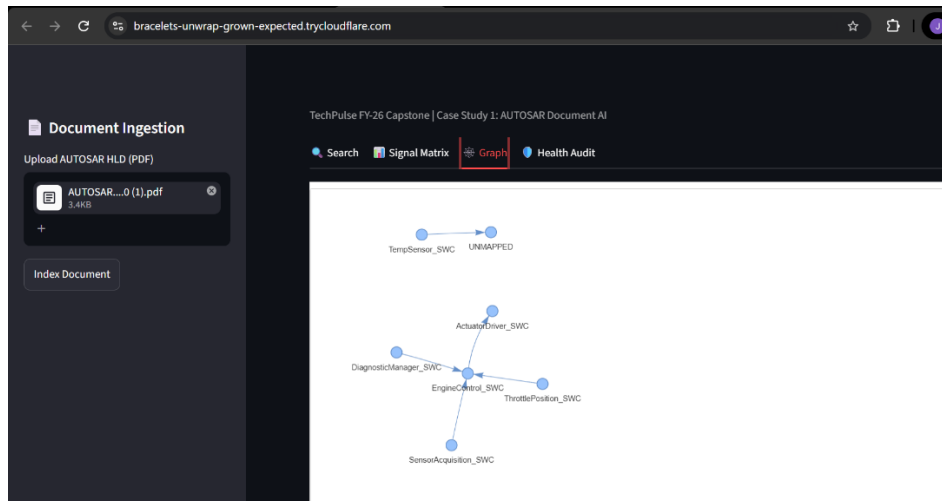


[Figure 1: Streamlit Dashboard — Grounded Search with Source Page Citation (Tab 1)]

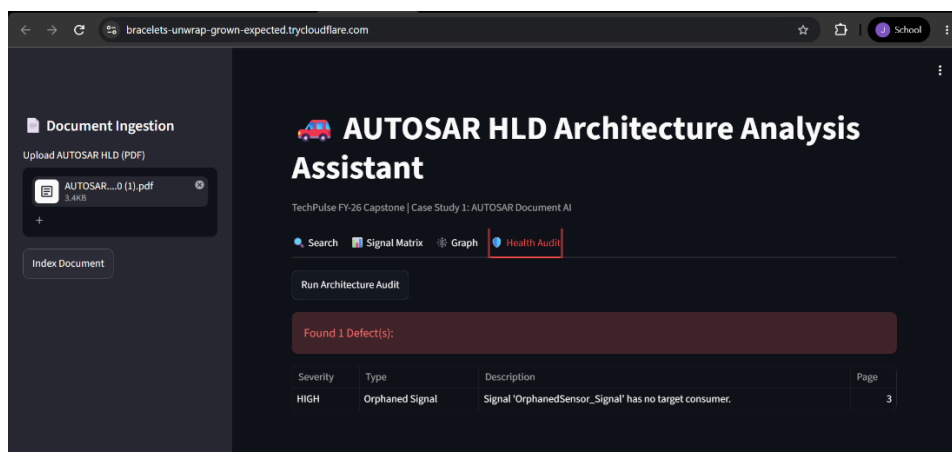
A screenshot of the same web application, but with the "Signal Matrix" tab selected. It displays a table titled "Table from Page 3" with 5 columns: Signal Name, Source SWC, Target SWC, Data Type, and Interface Type. The table contains 5 rows of signal mapping data.

	Signal Name	Source SWC	Target SWC	Data Type	Interface Type
0	ThrottlePosition_Signal	ThrottlePosition_SWC	EngineControl_SWC	float32	Sender-Receiver
1	EngineSpeed_Signal	SensorAcquisition_SWC	EngineControl_SWC	uint16	Sender-Receiver
2	FuelInjectionCmd_Signal	EngineControl_SWC	ActuatorDriver_SWC	float32	Sender-Receiver
3	OrphanedSensor_Signal	TempSensor_SWC	UNMAPPED	uint8	Sender-Receiver
4	DiagStatus_Signal	DiagnosticManager_SWC	EngineControl_SWC	boolean	Client-Server

[Figure 2: Extracted Signal Mapping Matrix with CSV Export Button (Tab 2)]



[Figure 3: Interactive Component Dependency Map Rendered via Pyvis (Tab 3)]



[Figure 4: Automated Architecture Health Audit Panel Flagging High Severity Defect (Tab 4)]

9. Reflection and Conclusions

1. Key Learnings: Gained practical experience in building layout-aware RAG pipelines, deploying local vector stores (ChromaDB), leveraging GraphRAG for visual software topology, and implementing deterministic rule engines for high-stakes automotive engineering applications.
2. Challenges Faced: Overcame local tunnel proxy errors and dynamic JavaScript module blocking in Google Colab by integrating Cloudflare Tunnels (cloudflared) and configuring absolute directory pathing inside Streamlit.

3. Limitations: The current parser is optimized for structured PDF specifications; parsing raw legacy scanned documents without native text layers would require an integrated OCR engine (e.g., Tesseract or PaddleOCR).
4. Future Improvements: Extend ingestion to process raw AUTOSAR XML (*.arxml) files directly, add multi-document traceability across multiple ECU sub-systems, and implement automated fix suggestion generators.

10. AI-Tool Usage Declaration

AI tool / model	Purpose	Artifact affected	Verified by Student (Y/N)
Google Gemini	Interactive code debugging, path resolution optimization, Cloudflare tunnel setup, and evaluation script generation.	Code/app.py, evaluate.py, Colab execution workflow	Y

11. Student Declaration

I confirm that this report represents my own understanding and contribution, that the work follows the approved case-study scope, and that I can explain every significant part of the implementation.

Signature:		Date:	06/10/2026
Name:	Joel Emmanuel	PRN:	123B1D022